

has been advocated by Nelson and others² has not been used as a source of light in this work. Most of the work recorded in this paper has been done with an inclosed type carbon-arc light operated at 220 volts d. c. at 13 amperes. Since the light is of the inclosed type the carbons can be operated with this current about 34 hours without renewal. The glass globe should be cleaned on each renewal of the electrodes.

Figure 1 shows (in comparison with sunlight) the spectrum of the arc used with the glass globe in position, taken on a panchromatic film in a quartz spectrograph. It is to be noted that the band spectra of both the arc and sun apparently become more continuous as the time of exposure is increased. Note also that while this arc spectrum shows much radiation in the moderate ultra-violet (350 to 400 $m\mu$) it shows none of the very short waves not found in sunlight but given by the quartz mercury arc. The electrodes used are of solid carbon 12.7 mm in diameter. The glass globe is opaque to ultra-violet wave lengths less than 320 $m\mu$ and infra-red radiation longer than about 4,500 $m\mu$. The total radiation at a distance of 38 cm (15 inches) from the center and at right angles to the arc on 12.6 amperes is 0.075 watt per cm^2 . This is distributed in the spectrum as shown in Table 1.

TABLE 1.—Spectral distribution of carbon-arc light with glass globe

Spectral range	Per cent of the total radiation
320 to 360 $m\mu$ -----	2.0
360 to 480 $m\mu$ -----	18.5
480 to 600 $m\mu$ -----	9.3
600 to 1,400 $m\mu$ -----	16.5
1,400 to 4,200 $m\mu$ -----	22.1
4,200 to 12,000 $m\mu$ -----	31.6

The large amount of infra-red of wave lengths 4,200 to 12,000 $m\mu$ emanates principally from the glass globe which becomes heated by the arc, and is different in spectral composition from the infra-red from the incandescent arc. Owing to unsteadiness of the arc, which travels around the electrode, the radiation measurements varied by 20 per cent. A steadier arc would be obtained by operating at 20 amperes, but at this current the carbons would be consumed too rapidly.

Figure 2 shows two light units. In the left-hand unit the light is shown in position in the center of the cylindrical sample holder, *H*.

² Proc. Am. Soc. Test. Mtls., **22**, Pt. II, p. 485; 1922; Proc. Am. Soc. Test. Mtls., **24**, Pt. II, p. 920; 1924; Proc. Am. Soc. Test. Mtls. **26**, Pt. II, p. 563; 1926.

In the right-hand unit the light, *L*, is shown raised out of the cylindrical holder.

3. RAIN

In addition to light exposure, the panels should be subjected from time to time to several hours of vigorous spraying with warm (100° F.) water. The panels can be removed to a special chamber for this purpose, but it is more convenient to produce this artificial rain in the chamber used for the light exposure. For this purpose the light is raised out of the cylinder, a common rotating lawn sprinkler put in, the bottom of the bracket supporting the light is protected from spray by a brass cylinder and the top of the cylinder holding the samples covered by galvanized iron covers.

Figure 3 shows the arrangement for spraying the panels with water. The right-hand unit with the cover removed shows the rotating spray, the slots containing test panels, and the protecting tube over the lower part of the lamp support. The left-hand unit shows the lid in place.

In order to simulate a hot, humid climate a fixed water spray has been mounted in the tank in such a manner that it may function while the lights are operating. Connections for this spray are made at the tee above and between the two tanks. (See fig. 3.)

The light cylinders are also now made to rotate slowly, at the rate of three revolutions per hour. Thus with the lights in operation and in position (see fig. 2), a gentle spray can wet the test surfaces periodically while exposed to the light.

4. TEMPERATURE CHANGES (REFRIGERATION)

An ammonia coil about 27 by 47 by 40 cm in an insulated chamber that can be cooled to -25° C. (-13° F.) is used to quickly chill the test panels.

The test panels on removal from the light chamber are placed on the rack shown, and the rack with samples is placed in the refrigerator and left there for one hour and then removed.

5. GASES

It is most convenient to transfer the panels to a separate cabinet for exposure to various gas mixtures. Such a cabinet for exposure to ozonized air is shown in Figure 4. Air is forced, by means of the small motor-driven fan shown in the lower left corner, through a silica gel dehydrator, then through a calcium chloride bottle, which serves to indicate that the silica gel is working efficiently, to the ozonizing apparatus and thence to the glass chamber on the right for holding the panels. Some water is placed in the bottom of this

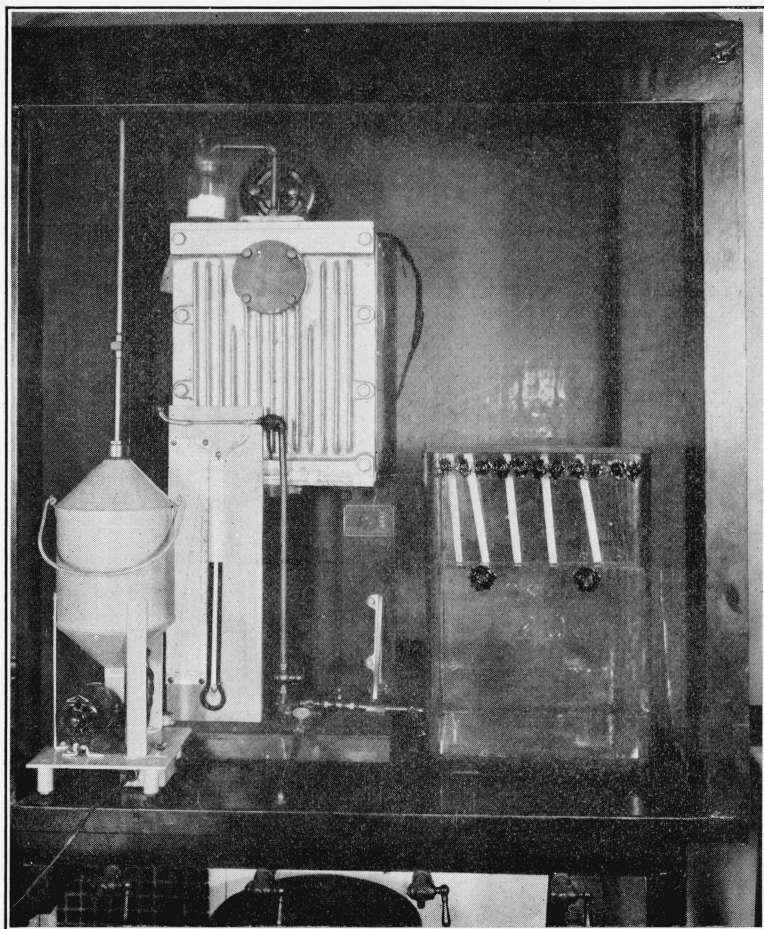


FIG. 4.—Cabinet for exposing to ozonized air

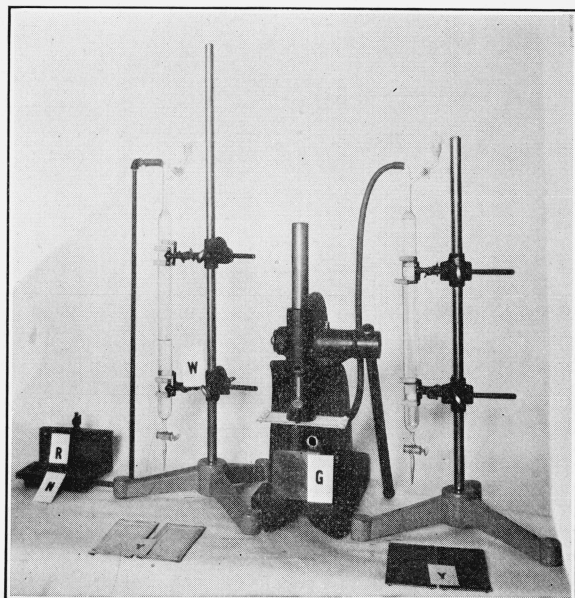


FIG. 5.—Apparatus for determining breakdown of film by passing air through it

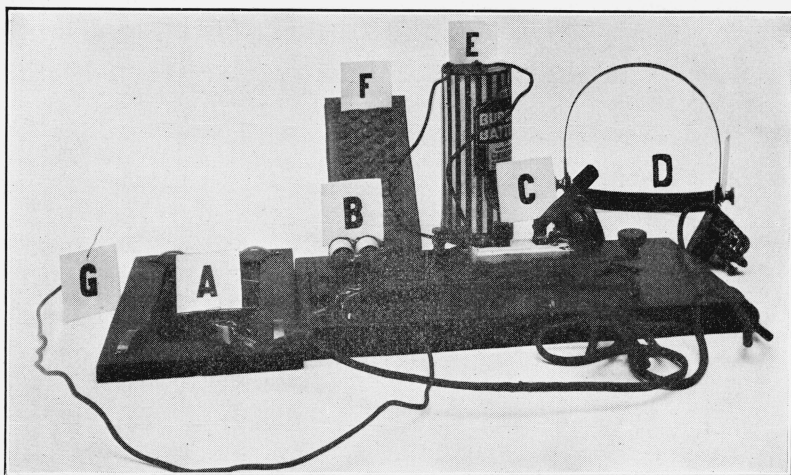


FIG. 6.—Photograph and wiring diagram of Wilson's apparatus for determining end point of paint failures

A, Steel wool contact frame. B, Buzzer. C, Switch. D, Head phones. E, Dry cell. F, Rubber stamp. G, Platinum tip

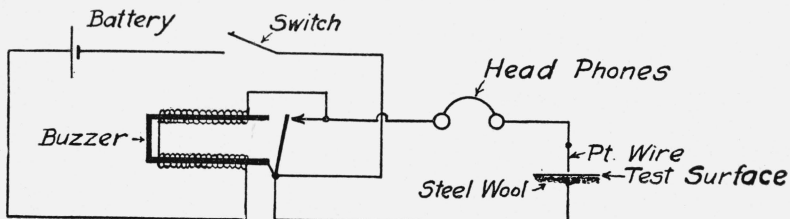


FIG. 7.—Wiring diagram

glass chamber to moisten the ozonized air. By suitable manipulation of the stopcocks in the pipe line the rate of flow can be determined by the flow meter, shown to the right of the silica gel holder, and samples for determining the ozone concentration can be taken through the pipe projecting toward the front. This equipment delivers about 660 liters per hour of air containing about 0.08 per cent of ozone by volume.

6. METHOD OF PREPARING TEST PANELS

It is a waste of time to make tests on single coats of protective coatings. This can be readily demonstrated by brushing and flowing paints on glass and, after drying, examining by transmitted light. It is advisable to always use at least two and generally three coats on the exposed test surface.

When using wood or metal, the back and edges of the panels are given three coats of aluminum paint (25 g of polished aluminum powder to 100 ml of long-oil water-resisting spar varnish). This aluminum paint is very waterproof and gives excellent protection to both wood and steel. Since, during exposure, it is not exposed to light it may be relied upon to last longer than any material being tested on the exposed surface.

In addition to tests on wood or metal, it is advisable to apply the material to 100-mesh wire gauze or to cotton cloth. This will be discussed later in connection with methods of determining the extent to which disintegration has taken place.

7. DESCRIPTION OF EXPOSURE CYCLE

Since light is the most important destructive agency in exposure tests, it seemed necessary to make use of as much light as possible. Therefore, the night hours (17 hours) were given over to light exposure. The idea in subjecting the panels to air containing a small amount (0.08 per cent) of ozone was to accelerate surface oxidation. In using refrigeration, the desired factor seemed to be a rapid change in temperature, rather than any definite time of exposure to the low temperature. The refrigerating unit was not assembled until some time after the light-exposure cylinder and gas chamber were made. In some of the earlier tests the panels were, therefore, not exposed to sudden temperature changes. The following tentative schedule shows the weathering cycle now in use:

Monday:

Ozonized air (wet atmosphere).....	3½ hours (9.30 to 1).
Water (tap water at about 100° F., from a rotating spray)	3 hours (1 to 4).
Light (carbon arc).....	17 hours (4 to 9).

Tuesday:

Refrigeration (-10° F.)	1 hour (9.30 to 10.30).
Ozonized air	2½ hours (10.30 to 1).
Water	3 hours (1 to 4).
Light	17 hours (4 to 9).

Wednesday:

Water	3 hours (9.30 to 12.30).
Refrigeration	1 hour (12.30 to 1.30).
Ozonized air	2½ hours (1.30 to 4).
Light	17 hours (4 to 9).

Thursday:

Refrigeration (-10° F.)	1 hour (9.30 to 10.30).
Ozonized air	2½ hours (10.30 to 1).
Light	3 hours (1 to 4).
Water	17 hours (4 to 9).

Friday:

Refrigeration (-10° F.)	1 hour (9.30 to 10.30).
Ozonized air	2½ hours (10.30 to 1).
Water	3 hours (1 to 4).
Light	17 hours (4 to 9).

Saturday:

Ozonized air	3½ hours (9.30 to 1).
Water	3 hours (1 to 4).
Light	17 hours (4 to 9 Sunday).

Sunday:

Light	24 hours (9 to 9 Monday).
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Total operating hours=165 divided thus:

	Hours	Per cent
Refrigeration	4	2.4
Ozonized air	17	10.3
Water	32	19.4
Light	112	67.9
Total	165	100.0

It will be noted that in this schedule only 165 out of 168 hours are accounted for. The remaining three hours are used for the inspection of the panels.

III. METHODS OF DETERMINING EXTENT TO WHICH DISINTEGRATION HAS TAKEN PLACE

1. OBSERVATION BY EXPERTS

Competent experienced observers frequently draw widely varying conclusions from inspection of the same panels, and the same observers draw varying conclusions at different times. This is well illustrated by the following:

One of the most carefully executed and widely known series of exposure tests was the Havre de Grace Bridge test of the American Society for Testing Materials. Nineteen paints designed as protective coatings for steel were used. Each of these paints was applied in three spreading rates—600, 900, and 1,200 square feet per

gallon. Triplicate panels were used for each spreading rate, making nine panels for each paint. The panels were exposed in November, 1906. The work was done with the utmost care and the various inspections were made by competent men experienced in such work. The divergent opinions of competent observers mentioned above are well illustrated in the following selection of 5 of these 19 paints.³ The inspection was made May 5, 1910. The examination of the panels was divided into three heads, viz:

TABLE 2.—*Ratings of 5 out of 19 paints inspected May 5, 1910*

Paint	Exam- ined for—	Aiken	Mac- Naughton	Tassin	Thomp- son	Hume	Average	Maxi- mum	Mini- mum
2-----	{ A-----	8	9	¹ 1	8	9	8.5	9	1
	{ B-----	10	6	¹ 1	8	9	8.2	10	1
	{ C-----	6.44	6	¹ 1	4.33	4.66	5.3	6.44	1
10-----	{ A-----	8	9	10	6	9	8.4	10	6
	{ B-----	9	6	10	6	9	8.0	10	6
	{ C-----	10	9	10	8	9	9.2	10	8
11-----	{ A-----	9	9	10	6	7	8.2	10	6
	{ B-----	9	6	10	7.14	9.66	8.4	10	6
	{ C-----	10	8	10	7.14	9	8.8	10	7.14
13-----	{ A-----	8	10	10	8	9	9	10	8
	{ B-----	10	10	10	10	10	10	10	10
	{ C-----	8.55	7	10	6	9	8.1	10	6
14-----	{ A-----	9	9	10	8	8	8.8	10	8
	{ B-----	10	9	10	10	10	9.8	10	9
	{ C-----	10	8	10	6	9	8.6	10	6

¹ These values were not considered in computing the average of paint No. 2.

A. Chalking.

B. Checking, cracking, "alligatoring," etc.

C. General surface condition, with a marking for each as follows:
Excellent, 10 to 8.

Good, 8 to 6.

Fair, 6 to 4.

Poor, 4 to 2.

Failure, 2 to 0.

As far as possible members of the subcommittee were requested to pair off with such other members of Committee D-1 who were present though not of the subcommittee on inspection. Those of the subcommittee present were *W. A. Aiken, chairman, *G. W. Thompson, *M. MacNaughton, *A. P. Hume, Anderson Polk, A. H. Sabin, *Wirt Tassin. Those marked with an asterisk (*) sent in individual reports. A. H. Sabin was the only one who collaborated

³ Taken from Proc. A. S. T. M., 10, pp. 105-106; 1910.

with any other member of the subcommittee; he was with the chairman, who also was accompanied by Mr. Gibboney, of Committee D-1. The individual marks were averaged from all nine panels, covering three rates of spreading.

Another lot of 51 panels was inspected by a committee of seven members on April 15, 1910,⁴ and by a committee of four on June 28, 1911.⁵ Three of the four members of the 1911 committee were also members of the 1910 committee. In the 1910 ratings 18 panels had a maximum rating of 10 and only 6 of these a minimum rating as high as 8 and 8 had a minimum rating of 6 or less. Sixteen panels had a maximum rating of 9, with only 4 of these having a minimum rating of 8, and 8 having a minimum rating of 5 or less. The variation in the remaining 17 panels is proportionally as great, with the exception of 2 which were marked "0" by all members of the committee. Table 3 showing results on 7 of these panels was compiled from the two reports.

TABLE 3.—*Ratings of 7 out of 51 paints*

Paint No.	Date of inspection	W. H. Walker	P. H. Walker	Gardner	Chapman
7-----	{ Apr. 15, 1910	----- 3	9	8	8
	{ June 28, 1911		4	5	7
24-----	{ Apr. 15, 1910	----- 5	5	5	8
	{ June 28, 1911		7	2	8
333-----	{ Apr. 15, 1910	----- 4	7	9	9
	{ June 28, 1911		4	7	3
555-----	{ Apr. 15, 1910	----- 9	9	8	9
	{ June 28, 1911		9	6	9
777-----	{ Apr. 15, 1910	----- 7	9	8	9
	{ June 28, 1911		10	5	7
100-----	{ Apr. 15, 1910	----- 5	10	7	9
	{ June 28, 1911		8. 5	4	8. 5
5555-----	{ Apr. 15, 1910	----- 4	1	0	3
	{ June 28, 1911		8	2	7

Not only does this table show wide variation of opinion among competent observers examining the same panels, but it will be noted that all of the three observers who took part in both inspections rated one paint higher and one of the three rated two other paints higher in 1911 than in 1910.

These and other similar examples convince the writers that one of the greatest needs for the study of paint and similar coatings is a

⁴ A. S. T. M. Proceedings, 10, p. 73; 1910.

⁵ A. S. T. M. Proceedings, 11, p. 192; 1911.

method or methods for determining the time of breakdown of the coatings.

2. PERMEABILITY TO WATER VAPOR

Muckenfuss⁶ more than 14 years ago suggested determining permeability to water as a measure of the breakdown of paint coatings and described preliminary experiments. Dunlap⁷ has described a method used at the Forest Products Laboratory for determining the breakdown on weathering of paint on wood.

While using the same general idea of these investigations—determining the progressive change in permeability on weathering—the writers attacked the problem in different ways.

Twenty-three No. 100⁸ wire sieves 95 mm ($3\frac{3}{4}$ inches) in diameter (71.26 cm² area) were given three coats as follows:

Sieves Nos. 1, 2, and 3, 8-gallon rosin-tung oil varnish.

Sieves Nos. 4, 5, and 6, 30-gallon rosin-tung oil varnish.

Sieves Nos. 7, 8, and 9, 50-gallon rosin-tung oil varnish.

Sieves Nos. 10, 11, and 12, the above 8-gallon varnish plus aluminum (2 pounds per gallon of varnish).

Sieves Nos. 13, 14, and 15, the above 30-gallon varnish plus aluminum (2 pounds per gallon of varnish).

Sieves Nos. 16, 17, and 18, the above 50-gallon varnish plus aluminum (2 pounds per gallon of varnish).

Sieves Nos. 19, 20, and 21, white lead-linseed oil paint.

Sieves Nos. 22 and 23, lead-zinc linseed-oil paint, B. S. Circular No. 89.

One week was allowed for drying between coats, and 1 week after applying the last coat the sieves were fastened by rubber bands to glass petri dishes containing dry CaCl₂, and then weighed. A blank consisting of a petri dish with dry CaCl₂, fastened with rubber bands to another petri dish, was also weighed. The samples, with the blank, were then put in air saturated with water for 20 hours, then conditioned for 1 hour at 30° C. and 32 per cent relative humidity and weighed again. This was repeated each day for 2 weeks. The average gains corrected for the blank were taken as the initial gains.

The coated sieves were then removed from the CaCl₂ dishes, exposed to the weather at 45° to the vertical facing south for a week, again fastened to the petri dishes over CaCl₂, exposed to saturated air for 20 hours, conditioned for 2 hours, weighed, the weight corrected for the blank, removed from the dishes and again exposed out of doors. A similar sieve uncoated and exposed to the same high

⁶ J. Ind. and Eng. Chem., **5**, p. 535; July, 1913.

⁷ Ind. and Eng. Chem., **18**, p. 1230; December, 1926.

⁸ The openings in No. 100 wire sieve cloth are 0.149 mm; the diameter of the wire is 0.102 mm, and the area of the open space about 35 per cent of the total area.

humidity condition showed an average passage of about 7.4 g of water. The results are shown in Table 4.

TABLE 4.—Permeability to water vapor of coatings on wire mesh exposed to weather

Material (three coats)	8-gallon varnish	30-gallon varnish	50-gallon varnish	8-gallon varnish +Al	30-gallon varnish +Al	50-gallon varnish +Al	White lead- linseed oil paint	Lead- zinc paint F. S. B. Specifi- cation No. 10
Initial water absorbed through film								
.....grams.....	0.024	0.033	0.054	0.011	0.016	0.024	0.074	0.074
Less than 0.15 g water absorption								
through film after.....weeks.....	8	31	1 43	31	1 43	1 43	1 43	1 43
Less than 0.25 g water absorption								
through film after.....weeks.....	8	35	-----	1 43	-----	-----	-----	-----
Less than 0.50 g water absorption								
through film after.....weeks.....	8	39	-----	-----	-----	-----	-----	-----
Less than 1.00 g water absorption								
through film after.....weeks.....	8	1 43	-----	-----	-----	-----	-----	-----
Less than 1.50 g water absorption								
through film after.....weeks.....	10	-----	-----	-----	-----	-----	-----	-----
Less than 2.00 g water absorption								
through film after.....weeks.....	11	-----	-----	-----	-----	-----	-----	-----
Less than 2.50 g water absorption								
through film after.....weeks.....	12	-----	-----	-----	-----	-----	-----	-----
Less than 3.00 g water absorption								
through film after.....weeks.....	12	-----	-----	-----	-----	-----	-----	-----
More than 3.00 g water absorption								
through film after.....weeks.....	13	-----	-----	-----	-----	-----	-----	-----

¹ More than.

TABLE 5.—Permeability to moisture—accelerated cycle

Paint	Water absorbed by calcium chloride through coatings (grams)			
	Before exposure	After exposure to accelerated cycle		
		41 days	48 days	55 days
White lead and linseed oil-----	{ 0.03 .03	0.06 .06	0.07 .09	0.08 .06
Lithopone and linseed oil-----	{ .03 .03	.12 .11	.20 .25	.41 .48

The early stages of the above investigation were so encouraging that two paints were tested in a somewhat similar manner in the accelerated cycle (without refrigeration). For this test rectangular brass frames 7.5 by 15 cm to hold the No. 100 sieve cloth were prepared and three coats of the paint applied as before. Permeability was determined by clamping a glass weighing tube containing CaCl_2 to the cloth, using rubber washers to make tight joints. The results are shown in Table 5. Here again the results are what experience with these paints would indicate.

3. PERMEABILITY TO AIR

This work on sieve cloth was so encouraging that a number of similar panels were prepared, coated with various well-known paints, and subjected to the accelerated cycle. It was soon found, however, that the experimental errors were so large, owing to the small amount of water taken up and the difficulty of learning whether the joint between the film and the weighing tube was tight, that it was decided to abandon the gravimetric method and to use a method of measuring the air passing through the film under definite conditions. Apparatus for this had to be designed and made, and this took so much time that the exposure to the accelerated cycle had in many cases gone too far before any measurements were made.

Figure 5 is a photograph which shows the apparatus finally developed. *Y* shows unpainted and painted wire sieve cloth in brass frames; *N* and *R* show bottom and top of a cell for holding the whole frame and to the left of *G* a similar cell for clamping a portion of the painted gauze. Rubber gaskets on top and bottom of the gauze panels serve to make an air-tight connection when the cell is compressed in the arbor press *G*. The lower compartment in the cell is joined by an air-tight connection to one of the large burettes shown to the right of *G* and the left of *W*. The water is adjusted to a mark in the upper constricted portion and with the cock in the upper portion of the cell closed the burette cock is opened. When water does not fall in two minutes below a zero mark on the burette (previously determined), the joint is considered tight. The cock in the upper part of the cell is then opened, the burette read after two minutes, and the volume recorded. The small cell has the disadvantage that at each test the painted gauze is compressed between rubber gaskets, and the advantage that three readings can be made on each panel. The large cell, *R* and *N* (which on account of the difference in volume gives a different zero point on the burette), with the bolted type of frame shown in the figure, is very difficult to make tight. With frames soldered on the edges instead of bolted, tight joints can be made with this cell. It has the advantage of not compressing the test film and the disadvantage of giving only one reading on each panel.

Since all of the paints were applied to the bolted panels, the small cell was used, and three readings were taken on each panel or six on each paint.

The exposures to the accelerated cycle were started July 19, 1927. After one month panels 7 and 8 (lithopone linseed-oil paint) showed such complete disintegration by visual inspection that they were

removed from the test and kept in a clean, dry container. The remaining panels were subjected to the accelerated cycle for four months. The permeability tests were made November 21, 1927. The results are shown in Table 6.

TABLE 6.—*Permeability of various paints after four months in accelerated cycle*

Panel No.	Paint exposed July 19, 1927	Average permeability (ml air in 2 minutes) on Nov. 21, 1927
5 and 6	Basic carbonate white lead and linseed oil	118.5
7 and 8 ¹	Lithopone and linseed oil	122.5
9 and 10	5 and 6 paint + lampblack	76
11 and 12	Red lead and linseed oil	119.5
15 and 16	Black linseed oil paint, F. S. B. specification No. 14b	1.5
17 and 18	Red enamel, F. S. B. specification No. 19b	1.5
19 and 20	Titanium zinc paint, F. S. B. specification No. 278	16
21 and 22	19 and 20 paint + lampblack	38
23 and 24	Green linseed oil paint, F. S. B. specification No. 15	25.5
25 and 26	Lead-zinc linseed oil paint, F. S. B. specification No. 10b	11
27 and 28	25 and 26 paint + lampblack	13
	Unpainted No. 100 sieve cloth	132

¹ These panels were exposed exactly 1 month and then discontinued, because under the microscope the coating was cracked all over. They were by far worse than any other panels.

In general, these results confirm the other experiments in indicating that the method is valuable. The lithopone paint which shows up worst in the test is known to be a paint of poor durability. Black linseed oil paint is known to be one of the most durable paints. The red enamel, which is the only paint equaling the black in this test, is also a very durable paint. The slightly better showing of the untinted titanium zinc and lead zinc paints as compared with the same paints tinted is not what would be expected. It is possible that longer exposure may show different results with these; for this reason these discrepancies are not considered of any moment. The decided breakdown of red lead as compared with two of the white paints suggests that the supporting surface may influence the results. It is possible that bronze wire screen has some deleterious effect on red lead-linseed oil paint. It is clear that more work must be done with this method of determining the end point of paint failures before a positive opinion as to its real value can be expressed. However, the results as a whole are encouraging, and we expect to repeat some of the work in Table 6, and since the nature of the wire gauze may influence the behavior of some paints it is proposed to apply paints to cotton cloth as well as to wire gauze.

4. ELECTRICAL CONDUCTIVITY TEST

On August 31, 1927, Doctor Liebrich and Doctor Kerscka, of the Chemisch-Technische Reichsanstalt, Berlin, showed one of the writers a simple apparatus for testing the continuity of an insulating film on metal. A telephone receiver is connected with the back (unpainted) side of the steel panel. A voltage from an interrupted source is placed across the receiver, and the return wire ending with a fine metal brush is drawn across the painted surface. Breaks in the paint surface are indicated by a buzz in the telephone.

J. H. Wilson, assistant chemist in this laboratory, has investigated this method. He found that the simple brush used by Liebrich and Kerscka would detect rather large uncovered areas but would not detect many of the visible cracks. Wilson has, however, modified the method by placing drops of a conducting liquid that will readily wet the coating on the painted surface and completing the circuit by dipping the end of the wire in these drops. The test panel, after some of the paint is scraped from the back side, is laid on a frame of metal wool, which is connected with one lead from the battery, and evenly spotted with the conducting liquid with the rubber stamp (*F* in fig. 6). After about two minutes the platinum tip of the other lead is inserted in each of the drops. Since there are 50 drops, the number of buzzes multiplied by two gives the percentage of breakdown.

Ten commercial samples of olive green automobile enamels (oleo-resinous type) were tested on 7.5 by 15 cm black iron panels. Three coats, allowing one week between coats, were applied to all panels, four panels being prepared with each sample. Two of these four panels were subjected to the accelerated cycle (without refrigeration) and two were exposed outdoors 45° to the vertical, facing south. The exposures were started in March, 1927. The accelerated tests were rated by inspection after 10 days. The outdoor tests were rated by inspection after one month. The accelerated tests were then continued for a total of 35 days and the outdoor tests for 146. After 35 days' accelerated testing each panel showed by inspection greater deterioration than the corresponding 146-day outdoor panel.

They were then tested using Wilson's method. A summary of the results is given in Table 7.

TABLE 7.—Tests of olive green oleoresinous automobile enamels

No.	Composite rating on loss of gloss, discoloration, chalking, and cracking after 10 days' accelerated test	Rating on cracking alone after 10 days' accelerated test. In the columns marked on scale of 10, 1 being best	Rating after one month outdoor exposure	Rating by Wilson's test, per cent destruction	
				Accelerated panels after 35 days' accelerated test and storing in clean, dry compartment	Outdoor panels after 146 days' exposure and storing in clean dry compartment
1	A	1	No change-----	20	2
2	B	1½	Film whitened-----	12	2
3	A	1	No change-----	34	2
4	C	7	Checked all over----	100	92
5	A	1	No change-----	14	2
6	B	6	-----do-----	52	6
7	A	4	-----do-----	14	0
8	C	9	Checked all over----	100	58
9	C	9	-----do-----	100	32
10	A	1	No change-----	40	2

A=One of five best.

B=Intermediate.

C=One of three worst.

This test and similar results obtained with various bituminous materials on aluminum indicate that this method is very useful for recording the failure of protective coatings on metal. The method is rapid and, with proper care, quite easy of execution. It is important, however, to select a suitable wetting liquid. For glossy, hard coatings that have a tendency to repel water a dilute (0.5 per cent) soap solution is generally suitable. On chalky, soft films this solution spreads so rapidly that the individual drops flow together before the contacts can be made. For such coatings a dilute (0.5 per cent) solution of sodium chloride is generally suitable.

This test can be applied to almost all organic coatings for steel. Aluminum paint, however, has been found to be a sufficiently good conductor to give a buzz even when there is no break in the film. Other such coatings may be found, hence one should make sure that the undamaged coating does not cause a buzz before using this test. As may be expected, the intensity of the buzz given by the different drops varies greatly. However, the method gave good results.

5. USE OF MILLIAMMETER

It seemed desirable to the authors to obtain some quantitative measure of the current passing through various sections of the paint coatings. With valuable assistance from C. M. Saeger, jr., the following methods were used: A milliammeter, having a scale reading from 0 to 60 milliamperes, was connected in series with a battery and the test specimen. The current was adjusted with a resistance